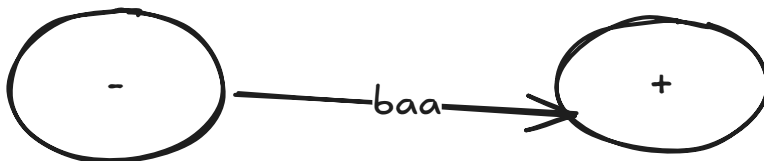


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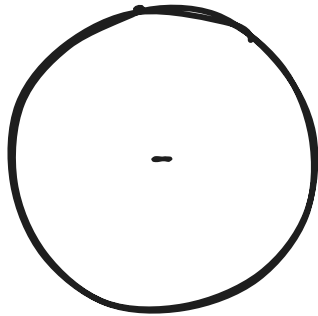
Transition Graph

- Difference between Finite Automata (FA) vs Transition Graph (TG)
 - FA had restrictions.
 - Every state must have transitions equal to the language characters.
 - We can read letters but not strings.
 - It is DFA (Deterministic Finite Automata).
 - No null string transition.
 - TG has lifted up most of the restrictions above
 - We can move from one state to other by reading strings.
 - Multiple outgoing transitions for a single character are possible.
 - Outgoing null transition is also possible.
 - It is NFA (Non-deterministic Finite Automata)
- Transition Graph Definition
 1. Finite number of state **at least one** of which is designated as start state. There may be many (maybe none) final state.
 2. An alphabet Σ of possible input characters $\Sigma = \{a, b\}$
 3. A transition that tells how to proceed from one state to another by reading some input character or string.
- Every FA is a TG, but not every TG is FA.

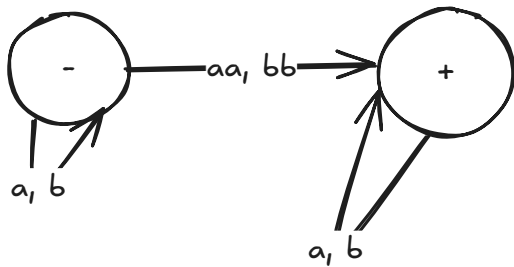
Build a TG which accepts 'baa'



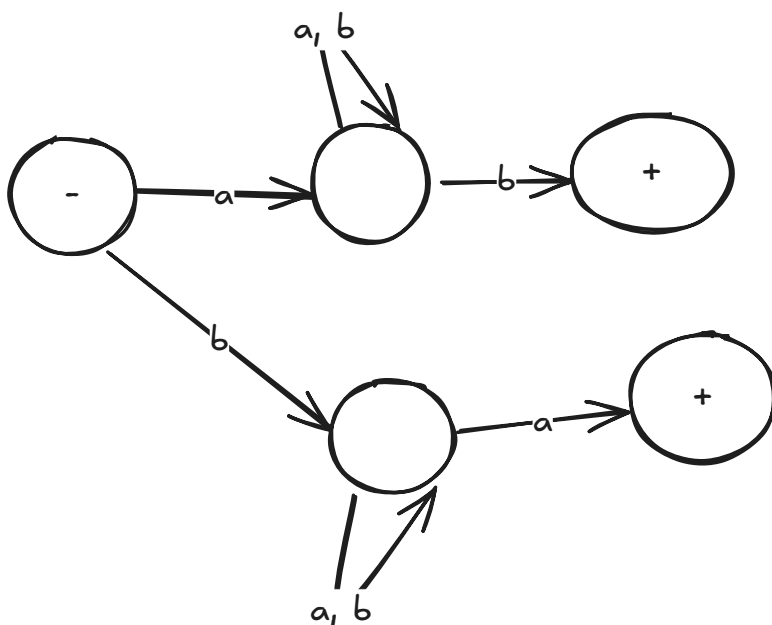
TG which accepts nothing



TG accepts double letters $(a+b)^*(aa+bb)(a+b)^*$



TG ending with different characters



Even-Even TG $\{aa + bb + (ab + ba)(aa+bb)^*(ab+ba)\}^*$

